Homework 4

ECON 470, Spring 2026

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1. Hospitals Filing Multiple Reports in the Same Year

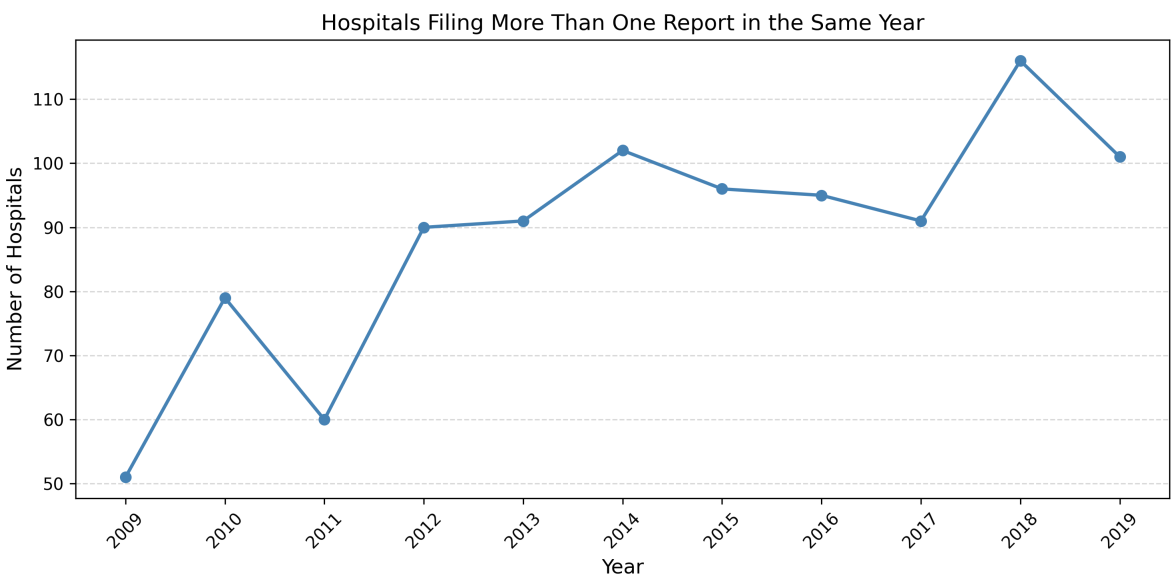

Figure 1: Number of hospitals filing more than one report per year, 2009–2019

Q1 answer: The number of hospitals filing more than one report in the same year ranges from a low of 51 in 2009 to a peak of 116 in 2018. Duplicate reports typically occur when a hospital changes its fiscal year and ends up with two cost reports falling in the same calendar year.

2. Unique Hospital IDs After Removing Duplicates

Q2 answer: After collapsing to one report per hospital per fiscal year, there are 6,860 unique Medicare provider numbers in the data across 2009–2019.

3. Distribution of Total Charges by Year

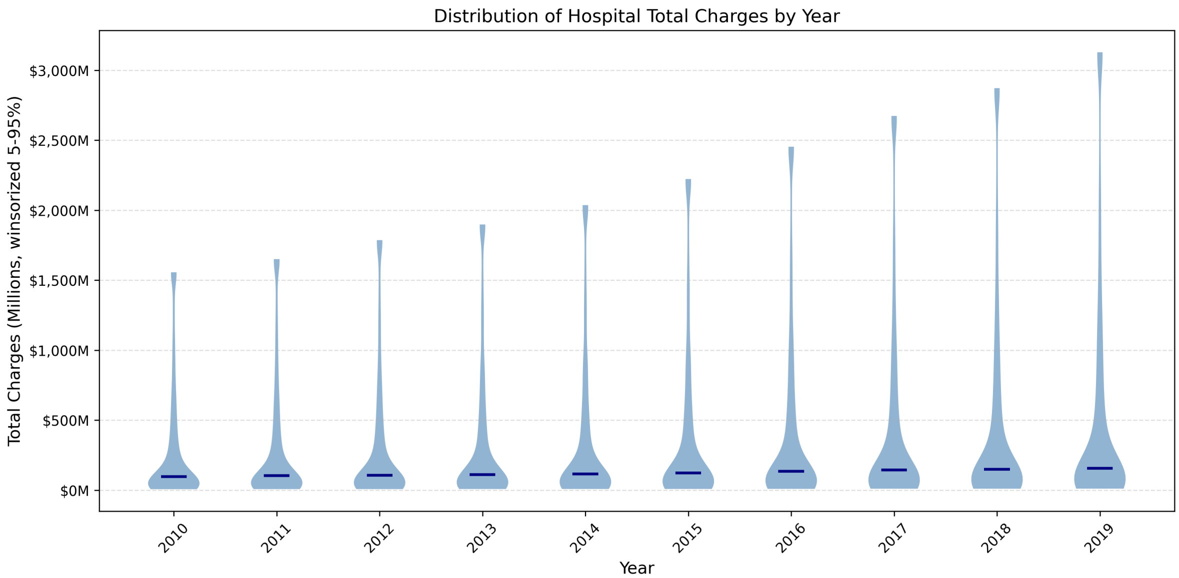

Figure 2: Distribution of total charges by year, winsorized at 5th and 95th percentiles

Q3 answer: Total charges rise steadily each year, with both the median and upper tail increasing. The distribution is right-skewed in every year, driven by a small number of large academic hospitals with very high charges.

4. Distribution of Estimated Prices by Year

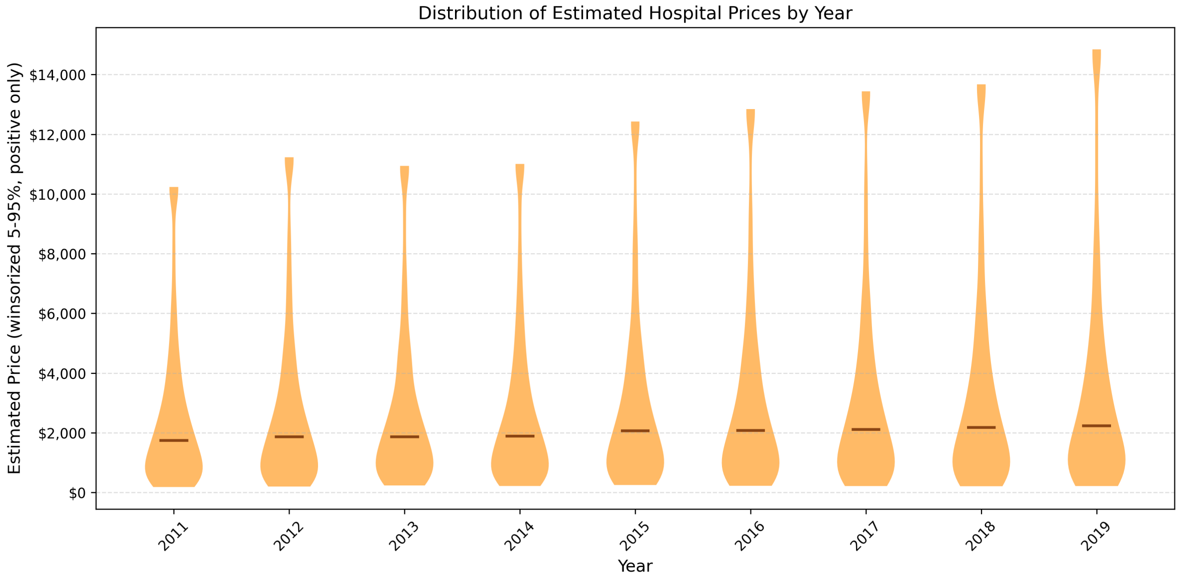

Figure 3: Distribution of estimated prices by year, winsorized at 5th and 95th percentiles, positive values only

Q4 answer: Estimated prices drift upward from 2010 to 2019, consistent with hospitals gaining leverage in commercial negotiations over time. The distributions are more symmetric than total charges since this measure nets out Medicare payments and scales by non-Medicare discharges.

5. Share of Hospitals Penalized Under HRRP and/or VBP

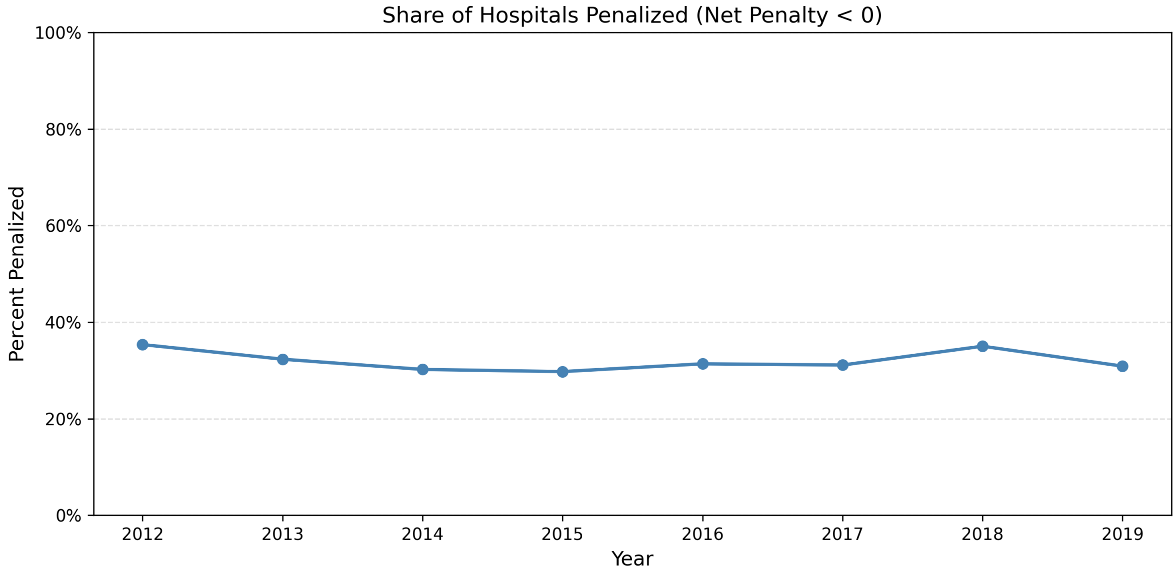

Figure 4: Share of hospitals penalized (net penalty < 0) under HRRP/VBP, 2012–2019

Q5 answer: About 35% of hospitals faced a net penalty in 2012, the first year HRRP took effect. The share rises to 32% in 2013 as VBP came online alongside HRRP, peaking at 35% in 2012. From there, the share levels off and ends at roughly 31% by 2019, suggesting most hospitals remained vulnerable to combined payment reductions throughout the later years of both programs.

6. OLS Estimates of Net Penalties on Price Changes

Table 1: OLS estimates of the effect of net penalties on price changes

	Spec (1)	Spec (2)	Spec (3)
Net Penalty ($1000s)	-0.034	-0.040	-0.044
	(0.025)	(0.025)	(0.024)
Avg Beds (pre-2012)		1.435	-0.261
		(0.457)	(0.620)
Avg Medicaid Discharges			0.021
			(0.007)
N	2,872	2,872	2,872
R^2	0.003	0.008	0.015

Q6 answer: Across all three OLS specifications the net penalty coefficient is small and negative, ranging from -0.034 in the simple specification to -0.044 with full controls, each estimated on 2,872 hospitals. Since more negative means a larger penalty, a negative coefficient indicates that more heavily penalized hospitals saw slightly smaller price increases on average. The R-squared values range from 0.003 to 0.015, meaning the penalty variable explains very little of the variation in price changes. OLS is probably biased here because penalty assignment isn't random. Sicker patient populations tend to drive both higher penalties and lower commercial pricing power.

7. Net Penalty vs Pre-2012 Medicare Discharges

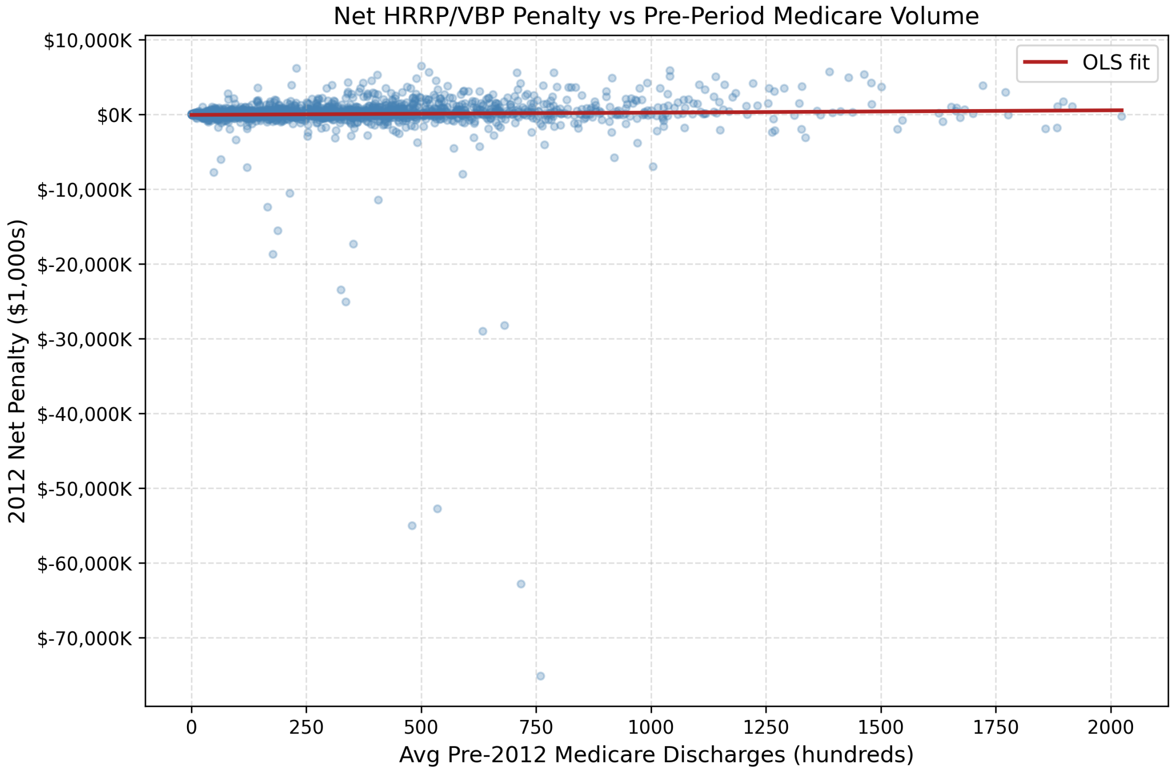

Figure 5: Net HRRP/VBP penalty vs average pre-2012 Medicare discharges (hundreds)

8. First Stage and Reduced Form

Table 2: First stage: net penalty on pre-2012 Medicare discharges

	Spec (1)	Spec (2)	Spec (3)
Avg Medicare Discharges (100s)	1.457	1.221	0.690
	(0.331)	(0.935)	(1.044)
Avg Beds (pre-2012)		0.974	0.334
		(2.670)	(2.625)
Avg Medicaid Discharges			0.030
			(0.013)
N	2,872	2,872	2,872
R^2	0.023	0.023	0.027
F-stat	19.4	28.1	22.9

Table 3: Reduced form: price change on pre-2012 Medicare discharges

	Spec (1)	Spec (2)	Spec (3)
Avg Medicare Discharges (100s)	0.446	0.780	0.486
	(0.135)	(0.302)	(0.309)
Avg Beds (pre-2012)		-1.374	-1.729
		(0.967)	(0.978)
Avg Medicaid Discharges			0.017
			(0.007)
N	2,872	2,872	2,872
R^2	0.006	0.007	0.011

Q8 answer: The first stage shows that hospitals with more pre-2012 Medicare discharges received larger net penalties, with a coefficient of 1.457 per 100 additional discharges. The F-statistic of 19.4 well exceeds the normal threshold of 10, confirming that pre-period Medicare volume is a relevant instrument for penalty assignment. The reduced form coefficient of 0.446 (SE 0.135) shows that hospitals with higher Medicare volume also saw larger price increases, which is the identifying variation the IV strategy uses to recover a causal estimate.

9. IV Estimates of Net Penalties on Price Changes

Table 4: IV estimates of net penalties on price changes

	Spec (1)	Spec (2)	Spec (3)
Net Penalty ($1000s)	0.306	0.639	0.704
	(0.121)	(0.588)	(1.248)
Avg Beds (pre-2012)		-1.997	-1.964
		(2.644)	(2.571)
Avg Medicaid Discharges			-0.005
			(0.044)
N	2,872	2,872	2,872

Q9 answer: The IV estimate of the net penalty coefficient is 0.306 (SE 0.121) in the simple specification and 0.704 (SE 1.248) with full controls, compared to the OLS estimate of -0.034. The IV flips the sign, suggesting that once penalty assignment is instrumented by pre-period Medicare volume, hospitals facing larger penalties actually increased their commercial prices more. This reversal is consistent with negative selection in OLS. Hospitals penalized more heavily tend to serve sicker populations and have weaker commercial pricing power, making OLS downward biased. The standard errors widen considerably with added controls, reflecting the precision cost of the IV approach.

10. Local Average Treatment Effect

Q10 answer: The IV estimates identify a Local Average Treatment Effect for compliers. Hospitals whose penalty exposure was driven by their pre-2012 Medicare patient volume. This captures medium sized hospitals at the margin. They are large enough to face meaningful penalties as Medicare volume grew, but not so dominant that their penalty status was predetermined regardless of volume. Very small hospitals with too few Medicare patients to be penalized under any scenario, and very large academic centers that would be penalized regardless, are not identified by the instrument. The LATE could differ meaningfully from the average treatment effect if those larger safety-net hospitals respond more aggressively to penalties than the mid-sized compliers the instrument captures.